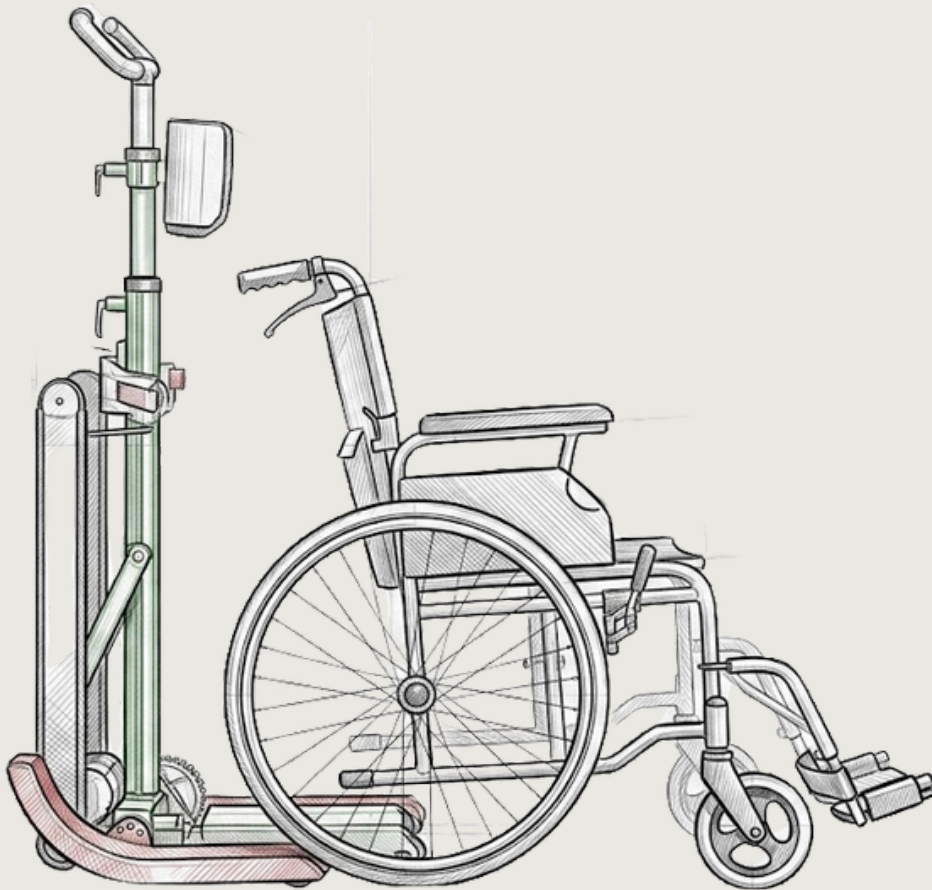


PROJECT PROPOSAL

A Generic Detachable Attachment for Existing Wheelchairs to Enable Moving Up Staircases



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1. PROJECT OVERVIEW

1.1 Title

A Generic, Detachable, Motorized Attachment for Standard Wheelchairs to Enable Staircase Navigation

1.2 Introduction

This project focuses on the conceptualization, design, and development of a motorized, detachable attachment engineered to retrofit standard manual wheelchairs, thereby granting them the ability to navigate staircases safely and efficiently. Designed by Team S.A.F.E., our solution is intended to bridge the critical gap between prohibitively expensive, fully integrated power-climbing chairs and affordable, yet functionally limited, manual mobility alternatives. By leveraging high-strength, low-density alloys and a sophisticated power-assisted drive mechanism, the attachment ensures a highly stable, horizontal seating experience for the user. Concurrently, it drastically minimizes the physical exertion required by caregivers, particularly during complex maneuvers such as U-turns and L-turns on narrow stair landings.

1.3 Project Purpose and Need

The necessity for this project stems from a fundamental failure in current mobility solutions to provide stair-climbing capabilities that are universally accessible, economically viable, and ergonomically safe. At present, a vast majority of wheelchair users and their caregivers are forced to rely on manual lifting to overcome architectural barriers. This archaic method poses significant safety risks, including the potential for severe falls, and places extreme physical and musculoskeletal strain on both the user and the helper.

This project addresses the urgent market demand for a "Universal Fit" solution. By incorporating a quick-clamp mechanism, the device will be fully compatible with a wide array of manual wheelchairs, offering a highly cost-effective alternative to expensive permanent building lifts or fully integrated power chairs.

Furthermore, there is a pronounced operational requirement for portability and agility. While existing stair-climbing systems (such as tracked climbing platforms) are notoriously heavy and bulky, our design philosophy prioritizes a lightweight assembly of **less than 10 kg**. Utilizing aerospace-grade alloys and a specialized mechanism that powers tire rotation via an independent motor, the unit is specifically engineered to navigate tight U-turns and L-turns in restrictive residential environments.

Crucially, the project fulfills a technical gap in user ergonomics. Unlike competing devices that

tilt the user backwards at severe angles, our design provides a stable, horizontal seating position throughout the climb. This is coupled with an integrated, automatic fail-safe braking system to ensure maximum security on steep inclines. Ultimately, the design is optimized for manufacturing feasibility using standardized components to ensure market viability, while embracing environmental stewardship through the use of recyclable materials and an energy-efficient powertrain.

2. ANALYSIS OF EXISTING SYSTEMS

The current market for stair-climbing accessibility is primarily bifurcated into permanent structural infrastructure and mobile platforms. The following analysis benchmarks these conventional systems against our project's core requirements for portability, maneuverability, and cost-effectiveness.

2.1 Fixed Building Lifts

Specifications: These encompass permanent mechanical installations such as elevators, platform stair-lifts, and vertical platform lifts. They provide high-capacity vertical transport with integrated safety controls and automated motorized operation.

Limitations: Despite offering high safety standards and ease of use, they inherently lack portability and are strictly fixed to a single location. Furthermore, the capital required for installation is often prohibitively expensive, and architectural constraints frequently make them impossible to retroactively integrate into older buildings or residential environments with limited spatial footprints.

2.2 Tracked Climbing Units

Specifications: These are heavy-duty, mobile, motorized units that utilize continuous rubber tracks to "crawl" up and down staircases. A standard wheelchair is typically secured onto the unit's platform before operation.

Limitations: Tracked units are notoriously bulky and heavy, severely compromising maneuverability. Performance benchmarking indicates that they struggle significantly with tight U-turns and L-turns on standard staircases. Their exorbitant cost and the immense physical effort required just to transport the unit limit their viability for daily, independent, or spontaneous use.

2.3 Manual Lifting (The Current Default)

Specifications: This process requires one or more helpers to manually lift and carry the wheelchair and its occupant over stairs and obstacles. It requires no specialized hardware or financial investment.

Limitations: Manual lifting introduces the highest degree of safety risk, lacking any fail-safe mechanisms to prevent catastrophic falls. It inflicts severe physical strain on caregivers, leading to long-term injury risks. Additionally, it subjects the wheelchair user to a high degree of instability, discomfort, and a loss of dignity.

2.4 Detachable Units

Specifications: These modular, power-assisted assemblies utilize clamping mechanisms to retrofit existing manual wheelchairs with stair-climbing capabilities. They typically feature motorized drive units intended to provide vertical elevation while reducing caregiver exertion.

Limitations: While more portable than fixed lifts, current detachable units in the market remain excessively heavy and bulky, hindering independent transport and attachment. Benchmarking reveals that these systems frequently exhibit poor maneuverability on narrow staircases, largely failing to clear tight landings. Furthermore, high manufacturing costs often render them financially inaccessible to the average consumer compared to standard mobility aids.

3. PROJECT OBJECTIVES & METHODOLOGY

3.1 Objectives

The primary objective of this project is to conceptualize, engineer, and prototype a modular, motorized attachment that seamlessly transforms a standard manual wheelchair into a highly capable stair-climbing vehicle. The specific technical objectives include:

1. **Universal Horizontal Adaptability:** To design a structural frame featuring a horizontally adjustable attachment mechanism, ensuring universal compatibility with manual wheelchairs of varying seat widths.
2. **Variable Staircase Synchronization:** To implement vertically adjustable tire positions, allowing the climbing mechanism to autonomously adapt to diverse tread and riser dimensions across varying staircase geometries.
3. **Lightweight Portability:** To achieve a total assembly weight of less than 10 kg by exclusively utilizing high-strength, low-density alloys, ensuring the device remains effortlessly transportable and deployable.
4. **Automated Safety Systems:** To engineer and integrate an automatic incline braking system that acts as an immediate fail-safe against electrical failures, mechanical faults, or loss of operator control on gradients.
5. **Enhanced Maneuverability:** To optimize the drive system architecture and frame geometry for navigating severe architectural constraints, specifically ensuring the capability to perform flawless U-turn and L-turn navigation on narrow stair landings.
6. **Operator Fatigue Reduction:** To deliver a fully power-assisted motorized mechanism that drastically minimizes the physical exertion currently required from caregivers during the ascent and descent phases.

3.2 Methodology

The project will execute a systematic, four-stage engineering methodology to guarantee technical rigor, safety, and commercial feasibility:

- **Concept & Definition:** A comprehensive literature and patent review of existing mobility solutions, followed by the formal definition of strict technical requirements for the universal frame, quick-clamp architecture, and adjustable mechanisms.
- **Design & Analysis:** Detailed 3D CAD modeling will dictate the adjustable frame architecture and the stair-climbing linkage mechanism. Rigorous Finite Element Analysis (FEA) will be conducted to validate the structural integrity, fatigue life, and stress distribution of the high-strength alloy frame under dynamic and static loads.
- **Prototyping & Testing:** Fabrication of a high-fidelity physical prototype utilizing standardized, off-the-shelf components to ensure future market viability. Bench testing will isolate and calibrate the drive logic. Subsequently, dynamic stair-climbing field tests will evaluate horizontal stability parameters and the responsiveness of the fail-safe braking system.
- **Evaluation & Final Documentation:** A comprehensive performance review will benchmark the prototype's turn radius, total weight, and battery efficiency against baseline data from existing manual and motorized methods, validating the fulfillment of all original project objectives.

3.3 Software Requirements

The following industry-standard software suites will be leveraged throughout the design, simulation, and analysis phases:

- **SolidWorks:** Utilized for advanced 3D CAD modeling, frame architecture layout, and Path-Mate motion simulations to accurately visualize and refine climbing kinematics.
- **ANSYS Mechanical:** Deployed for conducting comprehensive Finite Element Analysis (FEA) to verify structural safety margins and material integrity under simulated operational loads.
- **MATLAB:** Employed for the numerical analysis of powertrain efficiency, optimizing control algorithms, and calculating theoretical power consumption profiles per staircase climb.

4. PROJECT PLANNING & TIMELINE

4.1 Milestones

To ensure the timely and successful execution of the project, the following engineering milestones have been established:

- **Milestone 1:** Finalization of conceptual design and universal clamping mechanism requirements.
- **Milestone 2:** Completion of 3D CAD models and virtual kinematic motion simulations.
- **Milestone 3:** Successful validation of structural integrity via Finite Element Analysis (FEA).
- **Milestone 4:** Procurement of high-strength alloys, custom motors, and standard electronic components.
- **Milestone 5:** Successful fabrication and physical assembly of the Alpha Prototype.
- **Milestone 6:** Completion of dynamic field testing, load capacity verification, and braking system calibration.
- **Milestone 7:** Final design optimization, documentation, and project presentation.

4.2 Time Frame

The project lifecycle is divided into four distinct phases over a 14-week development schedule:

- **Phase 1: Research & Conceptualization (Weeks 1-2)** Literature review, defining parameters, and initial sketching.
- **Phase 2: CAD Design & FEA Simulation (Weeks 3-6)** Detailed SolidWorks modeling, ANSYS stress testing, and design revisions.
- **Phase 3: Procurement & Prototyping (Weeks 7-10)** Sourcing materials, machining alloy components, and assembling the drive mechanism.
- **Phase 4: Testing, Refinement, & Final Review (Weeks 11-14)** Bench testing, live staircase trials, MATLAB efficiency calculations, and final report generation.

5. IMPACT ANALYSIS

5.1 Economic Considerations

The economic viability of the proposed attachment is anchored in its potential to serve as a high-value, low-cost alternative to exorbitant capital infrastructure.

- **Manufacturing Feasibility:** The design is strictly optimized for production using standard mechanical components and globally sourced high-strength alloys. This approach suppresses manufacturing overhead, ensuring the final product reaches the mass market at an accessible price point.
- **Infrastructure Cost-Reduction:** By delivering a universal, mobile solution, this project negates the need for expensive permanent building modifications (e.g., fixed platform elevators), which are frequently financially out of reach for residential users and small-scale commercial entities.
- **Universal Compatibility:** The proprietary "Universal Fit" quick-clamp architecture ensures users can retain their existing manual wheelchairs. By averting the need to purchase specialized, fully integrated power chairs, the total cost of ownership for staircase mobility is drastically reduced.

5.2 Social Considerations

The societal impact of this project centers on radically improving the quality of life, autonomy, and safety for both wheelchair users and their dedicated caregivers.

- **End-User Safety and Stability:** The system is explicitly engineered with an integrated, automatic incline braking system, acting as an uncompromising fail-safe against mechanical or electrical failures.
- **Ergonomics and Comfort:** The design strictly prioritizes user experience by maintaining a stable, horizontal seating posture during the entirety of the stair navigation process. This prevents the severe physical discomfort and psychological insecurity associated with being sharply tilted backward (a common flaw in existing systems).
- **Helper Welfare and Inclusion:** By offloading the mechanical work to a motorized powertrain, the attachment drastically mitigates helper fatigue and nullifies the severe musculoskeletal injury risks historically associated with manual lifting.
- **Enhanced Maneuverability:** The agile design is painstakingly optimized for tight U and L-turn navigation, directly empowering users to access residential and public environments that were previously classified as entirely inaccessible due to restrictive landing dimensions.

5.3 Environmental Considerations

Proactive environmental stewardship is deeply integrated into the core design philosophy via conscious material selection and intelligent power management.

- **Sustainable Material Selection:** The structural frame is fabricated entirely from lightweight, high-strength alloys that possess a high recyclability rate. This effectively mitigates the long-term material waste generated during the manufacturing and eventual end-of-life disposal phases.
- **Energy Efficiency:** The electrical powertrain and motor control units are fine-tuned for minimal battery consumption per climb. This high-efficiency operation ensures a significantly reduced carbon footprint during daily consumer use.
- **Material Optimization:** Through iterative Finite Element Analysis (FEA) and topological CAD optimization, material distribution is minimized without sacrificing structural integrity. Achieving an overall assembly weight of under 10 kg drastically reduces the raw material volume required for global production.